

Feasibility Study Report

Comprehensive Multi-Dimensional Evaluation: Technical, Economic, Operational & Legal Feasibility

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Executive Summary

This report documents the feasibility analysis conducted for the **AI-Based Household Appliances Rental Platform (RentAI)**. The evaluation assesses whether deploying an AI-driven, NoSQL subscription marketplace is technically viable on consumer-tier and enterprise cloud hardware, economically profitable, operationally sustainable, and legally compliant.

2. Technical Feasibility

2.1 Hardware Resource Budget & CPU Inference

A critical technical objective was ensuring the complete platform—including database, backend API, machine learning inference, and local LLM generative AI—operates stably on standard workstation hardware (8 GB to 16 GB RAM, Quad-Core CPU, no discrete GPU required).

Component	Engine / Runtime	RAM Footprint	CPU Overhead
Database Store	MongoDB 8.3 Community	~120 MB	< 2% idle
REST Backend	Django 6 + Gunicorn / Waitress	~95 MB	< 3% idle
Churn & Rec Engine	LightGBM / FunkSVD (Pickled)	~60 MB	< 1% during scoring
Local LLM Daemon	Ollama (`qwen2.5:0.5b` / `rentai-llm`)	~580 MB	~35% during 2s inference
Client Application	React 19 SPA (Vite Static)	~25 MB (Browser)	Negligible
Total System	Integrated Stack	< 1,000 MB RAM	Smooth on Quad-Core

3. Economic Feasibility & Cost-Benefit Analysis

3.1 Zero-Cloud AI Token Cost (Ollama vs OpenAI API)

Traditional customer service bots relying on external APIs (e.g. GPT-4o) incur continuous operational expenditure (OpEx) scaling at ~\$0.01 to \$0.03 per customer interaction. At 50,000 queries per month, cloud APIs cost \$1,500/month (\$18,000/year). By using optimized local quantizations (`qwen2.5:0.5b` and `rentai-llm:latest`), RentAI operates at **\$0.00 marginal inference cost**.

3.2 Break-Even & Financial Projections

Metric	Month 3	Month 6	Month 12
Active Appliance Leases	150	600	2,200
Gross Monthly Rental Revenue (GMV)	₹2,70,000	₹10,80,000	₹39,60,000
Platform Commission (18%)	₹48,600	₹1,94,400	₹7,12,800
Server & Operational Hosting	₹4,500	₹12,000	₹28,000
Net Platform Yield	₹44,100	₹1,82,400	₹6,84,800

4. Operational & Legal Feasibility

- Operational Viability:** Field logistics are streamlined using algorithmic dispatching. Return inspections and security deposit release workflows prevent escrow disputes.
- Legal & Regulatory Compliance:** Compliant with the Information Technology Act (2000), Consumer Protection (E-Commerce) Rules (2020), and Digital Personal Data Protection (DPDP) Act (2023). All user passwords use salted hashing (PBKDF2/SHA256).

Feasibility Sign-Off: The technical, economic, operational, and regulatory assessments confirm that RentAI is highly viable, financially sound, and ready for commercial scale.